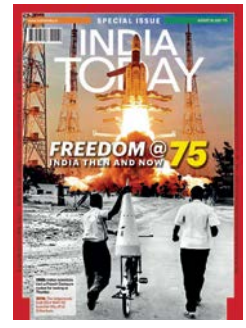
FROM THE

EDITOR-IN-CHIEF

INDIA@100

On the 76th Independence Day speech from the ramparts of the Red Fort, Prime Minister Narendra Modi declared: "During this Amrit Kaal (auspicious period), we must work hard to ensure that the dreams of 140 crore Indians become a reality and in 2047, when the tricolour is unfurled, the world is singing praises of a developed India." But the question is how we will realise the potential of this golden era in the quarter-century stretching to our 100th Independence Day. Undoubtedly, with a confluence of favourable international circumstances and forceful leadership, India is in a sweet spot to realise its full potential. The prime minister has given a clarion call with the words, "Perform, Reform and Transform."

In our Independence Day special issue, we have shortlisted 10 broad domains where change will have the maximum positive impact across the social and geographical spectrum. These will lay the grid on top of which more towering storeys—and uplifting stories—can be constructed. The Big Ten are: Energy, Computing, Communications, Health, Education, Agriculture, Transport, e-Commerce, e-Governance and Defence. These have been further subdivided into three components each where transformative change is already visible, and where we have the spark needed to get ahead of the global pack and define the path for ourselves. Alongside, we bring you profiles of the human agents of change in each category—scientists, innovators and visionaries at the very frontiers of a still-unfolding reality. It is of paramount importance that, alongside, we redouble efforts to create a healthy, skilled, knowledge-endowed populace that can activate this new potential. Taken together, what we present is a series of compelling tableaus that tell us how our lives can and will be transformed.



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We look at Energy, for instance, through a prism that splits it into three streams: green hydrogen energy, biofuels and battery technology. The first two have the capacity to unlock founts of energy that lie in abundance in our midst. And the third promises to get to the point where we can store monumental amounts of it. How deeply will this alter our realities? At the very root. The anxieties of recent decades were all about Peak Oil, the point at which all the world's known resources of petroleum would have been harnessed and the graph of availability would start dipping. This fear has driven terrible wars, and structured climate change negotiations in favour of the haves. But petroleum, the 'oil from rock', was the fuel of the 20th century. We are on the cusp of powering ourselves on subtler elements. For one of the world's fastest growing economies, the implications are far-reaching.

The same exciting potential attends to the fields of Computing and Communication. Artificial Intelligence is all over the news these days, but we have only yet seen the trailer. Similarly, Quantum Computing is at that launchpad stage where science was just before the atom was split. The whole buzzing ecosystem of knowledge around it is mostly monopolised by the West, but India is putting up the money to incubate genius. The Rs 6,000-crore National Quantum Mission (NQM) cleared by the Union cabinet in April is a step in that direction. Semiconductors are to computing what, in a slightly different way, battery technology is to energy. Luckily, India can pride itself on the edge it has in chip designing. It owes to the quality of our human resources that, over the past 20 years, several top global semiconductor design companies have set up centres here. As an industry player tells us, "We have 20-25 per cent of the world's chip designers in India." Expect the genetic code of our new economy to be written on such a triple helix.

In Communications, the next waves could be the size of a tsunami. With 6G, we will cross a threshold where human biology and data will be fused into one seamless continuum, with near-conscious networks reading our gestures and our virtual presence being summoned via holograms. If Satellite Internet will help us bypass our terrestrial challenges in universalising radiowave-based connectivity, LiFi will transcend radio and go to light itself to transmit signals. The payoffs in downstream fields like E-Governance or E-Commerce are easy to imagine. The first is already seeing something like a Gram Swaraj 2.0 on the back of a massive expansion of digital public infrastructure, which enables virtual documentation of everything and seamless delivery of public services. A similar revolution is sweeping the marketplace, propelled by novelties like drone-based delivery, contactless payments and open banking.

India's transition to developed country status will not be complete without a recoding of our 'software' part: our human resources. With technology spreading like light everywhere, Health and Education won't remain untouched. AI and advanced genetics-based medicine promise to spread wellness evenly in ways unthinkable before. And if machines are now capable of self-learning, they can also be aids to teaching: India's goal is to create an equitable STEM knowledge pool that can feed a virtuous cycle. On the farm front, we need a second Green Revolution, building degrees of sophistication and long-term vision into the sector. The urgent need to attain self-sufficiency in pulses and edible oils recalls the imbalances of old that we must avoid. There's another key component to the basics that need to be taken care: converting our vast physical spaces also into a buzzing network resembling the digital universe. The show-stopper here will be high-speed trains. That verve has already come to defence, where Make in India is proving to be quite a turbo-charged engine: the indigenous fighter jet Tejas will soon be powered by our own knowhow.

What you read in these pages dares us to reimagine every aspect of our lives. Hopefully, the rest of the century will not only be magnificent but will also carry an imprint that's our very own.



(Aroon Purie)